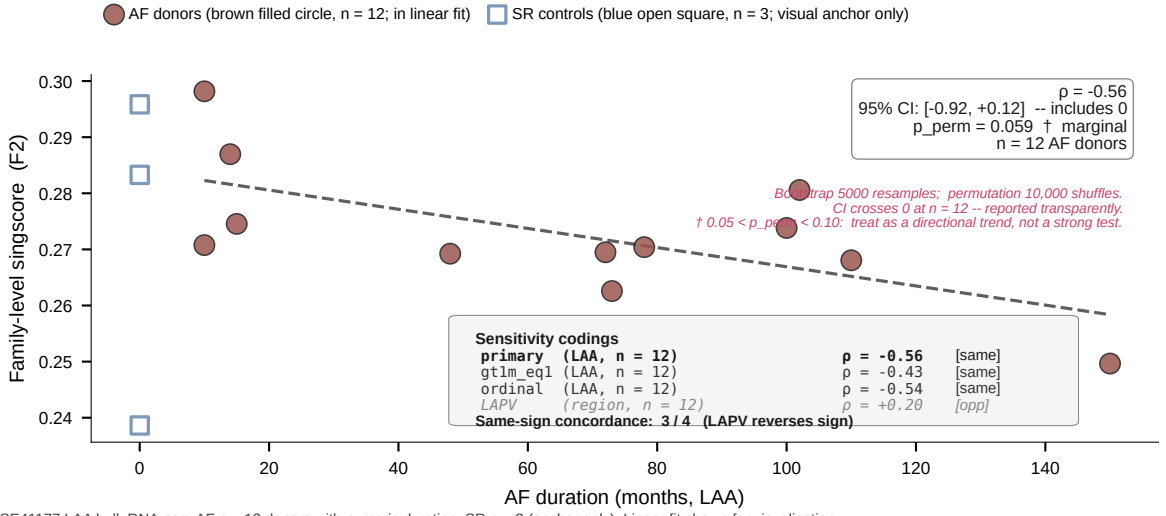
Fig 4e F2 Mito Ca² transport — Strong-trend (biological anchor)

Strong-trend (biological anchor)



GSE41177 LAA bulk RNA-seq. AF n = 12 donors with numeric duration; SR n = 3 (anchor only). Linear fit shown for visualisation only; ρ is rank-based. LAPV sensitivity row uses the paired LAPV-region cohort (n = 12), not LAA-pooled.

Data:

F2 (Mitochondrial Ca² transport, LA-biased baseline) shows the strongest single-family duration trend (ρ = -0.56, p_perm = 0.059, marginal). Three of four sensitivity codings agree in direction (3/4); the LAPV coding reverses sign (ρ = +0.20).

Literature anchor:

The negative direction matches the Ca² -handling remodelling of cardiomyocytes during sustained AF documented in Voigt et al. 2014 (Circulation), Heijman et al. 2014 (Circ Res) and Sun et al. 2018, supporting this trend as a genuine progression signal rather than noise.